

This guide is written for anyone who wants to self-study theoretical physics up to graduate qualifying exams and PhD candidacy. After that point you will have a much clearer idea of the direction you want to take. This is not a hard-and-fast rule, and not following it does not mean you will not succeed in theoretical physics. This is just advice. If you already own equivalent texts that are not mentioned here and you are confident they are a good alternative, please use them. Do not waste money.

Before you tackle science, and especially physics in any form, you should be comfortable with algebra, trigonometry, and the computational side of calculus, meaning the techniques of differentiation and integration. Rigorous calculus is not required at this stage and can be done alongside or after introductory physics. I recommend the following texts before introductory physics. Books marked (E) are extra: they are not necessary for the physics texts below, can be done alongside them, and will help later on.

- *Prealgebra* by Rusczyk
- *Algebra* by Gelfand, Shen; *The Method of Coordinates* by Gelfand, Glagoleva, Kirillov; *Functions and Graphs* by Gelfand, Glagoleva, Shnol; *Trigonometry* by Gelfand, Saul; *Geometry* by Gelfand, Alekseyevskaya; *Basic Mathematics* (E) by Serge Lang.
- *Plane Trigonometry I, II* by Loney; *Elements of Coordinate Geometry* by Loney
- *Geometry Revisited* (E) by Coxeter and Greitzer
- *Discrete Mathematics* (E) by Epp; *Introductory Discrete Mathematics* (E) by Balakrishnan; *Combinatorics* (E) by Mladenović; *Elementary Number Theory* (E) by Burton
- *How to Prove It* (E) by Velleman; *Book of Proof* (E) by Hammack
- *Introduction to Linear Algebra* by Lang; *About Vectors* by Banesh Hoffmann
- *Complex Numbers from A to Z* by Andreescu, Andrica
- *Precalculus* by Axler
- *Calculus: A Complete Course* by Adams, Essex
- *Elementary Differential Equations and Boundary Value Problems* by Boyce, DiPrima

Finally, you can start introductory physics. The undergraduate physics textbooks mentioned below require only the unmarked mathematics above. The marked texts will help later on.

- *An Introduction to Mechanics* by Kleppner, Kolenkow 2nd edition
- *Electricity and Magnetism* by Purcell, Morin
- *Vibrations and Waves* by French
- *Modern Physics* by Krane

After this we need to advance our mathematics. Coming from a mathematical physics background, I think I might overvalue mathematics, so those who do not wish to go down this path yet can refer to *Mathematical Methods in the Physical Sciences* by Boas, a text that covers all the mathematics an undergraduate physicist might need. Here is the undergraduate minimum for mathematics (speaking from my own experience at UAlberta):

- We start by proving the basic mathematics we have learnt. Do them in the order listed.
 - *Calculus* by Spivak and *Calculus Volume I* by Apostol
 - *Linear Algebra Done Right* by Axler and *Linear Algebra and Applications* by Strang
 - *Calculus Volume II* by Apostol and *Vector Calculus, Linear Algebra and Differential Forms* by Hubbard, Hubbard
- We go into analysis, differential equations, and abstract algebra. Do the books from left to right and top to bottom. You need not do all of them, just the topics you are interested in and need.
 - *Introduction to Real Analysis* by Bartle, Sherbert and *Analysis I* by Tao and *Understanding Analysis* by Abbott [Pretty tough, but take your time; it is worth it]
 - *Ordinary Differential Equations* by Arnold
 - *Abstract Algebra* by Dummit, Foote (*Abstract Algebra: An Introduction* by Hungerford and *Abstract Algebra* by Herstein) [Use Dummit, Foote and refer to the rest]
 - *Complex Variables and Applications* by Churchill, Brown and *Complex Analysis* by Ahlfors [Use one and refer to the other]
 - *Partial Differential Equations: An Introduction* by Strauss and *Applied Partial Differential Equations with Fourier Series and Boundary Value Problems* by Haberman [Use the second text for problems]
 - *Measure and Integral: An Introduction to Real Analysis* by Wheeden, Zygmund and *Real Mathematical Analysis* by Pugh [Royden, Fitzpatrick and Stein, Shakarchi and Tao Vol II supplement]
 - *Topology Part I* by Munkres and *Introduction to Topological Manifolds* by Lee
 - *Introductory Functional Analysis with Applications* by Kreyszig and *Functional Analysis* by Bühler, Salamon [*Applied Analysis* by Hunter, Nachtergaele helps you see the mathematical physics side]
- Now we complete the mathematics that will help with electives in theoretical physics.
 - *Calculus on Manifolds* by Spivak and *Calculus on Manifolds* by Munkres
 - *Introduction to Smooth Manifolds* by Lee
 - *Finite Group Theory* by Isaacs and *Representations and Characters of Groups* by James, Liebeck
 - *Topology: Part II* by Munkres and *Introduction to Topological Manifolds* by Lee

Finally, we can tackle the undergraduate core of physics, and I note after which math texts you can start the respective physics texts. Do them from top to bottom.

- *Classical Mechanics* by Taylor [Do Arnold alongside it]
- *Introduction to Electrodynamics* by Griffiths [Do Strauss alongside the chapters on Laplace's equation and separation of variables]
- *Thermal Physics* by Schroeder and *Fundamentals of Statistical and Thermal Physics* by Reif [After Hubbard, Hubbard text]

- *Introduction to Quantum Mechanics* by Griffiths and *Principles of Quantum Mechanics* by Shankar [Start with Griffiths after Arnold; start with Shankar after Churchill, Brown; the later chapters need the Functional Analysis texts]

This includes undergraduate electives that you should take if you have time. They are in no particular order.

- *Spacetime and Geometry* by Sean Carroll [Axler and Strang, Apostol, Hubbard, Hubbard, Arnold, Spivak or Munkres (Calculus on Manifolds)]
- *Introduction to Elementary Particles* by Griffiths and *Modern Particle Physics* by Thomson [Axler and Strang, Apostol (Volume II) and Hubbard, Hubbard, Arnold, Dummit, Foote (basic group theory, helpful for symmetries and $SU(2)$, $SU(3)$)]
- *The Oxford Solid State Basics* by Simon or *Introduction to Solid State Physics* by Kittel [Spivak and Apostol (Volume I), Axler and Strang, Apostol (Volume II) and Hubbard, Hubbard, Arnold, Strauss (Fourier series and wave equation chapters), Churchill, Brown (light use, for contour integrals and response functions)]
- *Introduction to Cosmology* by Ryden or *An Introduction to Modern Cosmology* by Liddle [Spivak and Apostol (Volume I), Axler and Strang, Apostol (Volume II) and Hubbard, Hubbard, Arnold]
- *Lie Algebras in Particle Physics* by Georgi and *Group Theory in a Nutshell for Physicists* by Zee [Axler and Strang, Dummit, Foote, Isaacs and James, Liebeck, Lee (Smooth Manifolds)]
- *Lectures on Quantum Field Theory* by Tong (*QFT in a Nutshell* by Zee, *QFT for the Gifted Amateur* by Lancaster, Blundell, or *A Modern Introduction to QFT* by Maggiore can be used as supplements) [Spivak and Apostol (Volume I), Axler and Strang, Apostol (Volume II) and Hubbard, Hubbard, Arnold, Churchill, Brown and Ahlfors, Strauss and Haberman, Dummit, Foote (groups and representations, for the Lorentz group), Kreyszig (Hilbert spaces and operators, for rigor)]
- *A First Course in String Theory* by Zwiebach [Spivak and Apostol (Volume I), Axler and Strang, Apostol (Volume II) and Hubbard, Hubbard, Arnold, Churchill, Brown, Strauss (wave equation, separation of variables, Fourier series), Dummit, Foote (helpful for the later chapters on D-branes and compactification)]
- *Geometrical Methods of Mathematical Physics* by Schutz or *The Geometry of Physics* by Frankel [Spivak and Apostol (Volume I), Axler and Strang, Apostol (Volume II) and Hubbard, Hubbard, Arnold, Dummit, Foote, Munkres (Topology Part I) and Lee (Topological Manifolds), Spivak and Munkres (Calculus on Manifolds), Lee (Smooth Manifolds); the last two are needed mainly for Frankel]

The mathematics you might need for the graduate core and beyond can be found in the following texts: *Mathematical Methods for Physicists* by Arfken, Weber, Harris; *Mathematics for Physics* by Stone, Goldbart; and *Mathematical Physics: A Modern Introduction to Its Foundations* by Hassani. The graduate core of physics includes these books from top to bottom:

- *Modern Classical Mechanics* by Helliwell, Sahakian and *Classical Mechanics* by Goldstein, Poole
- *Modern Electrodynamics* by Zangwill and *Classical Electrodynamics* by Jackson
- *Statistical Physics of Particles* by Kardar
- *Lectures on Quantum Mechanics* by Weinberg and *Modern Quantum Mechanics* by Sakurai 3rd edition
- *Course of Theoretical Physics* by Landau, Lifshitz

After completing the graduate core you will definitely have an idea of what you have to do. Choose your own path from here.